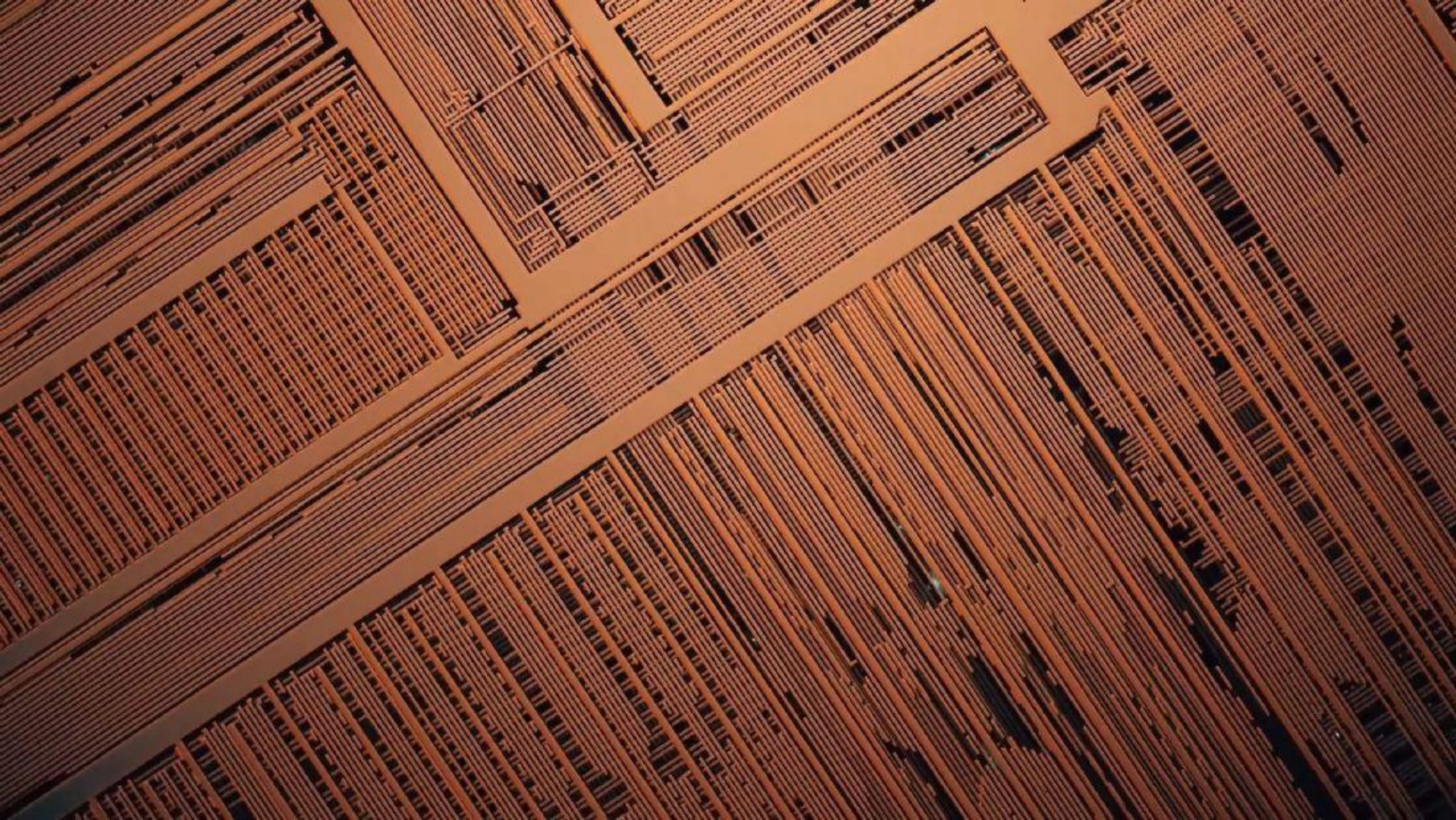
Firmguard

rehost • trigger • detect






STUXnet


NOTP
Ooops, you
follow the instructions!
Payment will be raised on
Your files will be lost on
Time Left
00:00:00:00
What Happened to
Can I Recover My Files?
How Do I Pay?
Contact Us
bitcoin
Check Payment
HACKING
XZ Utils

What's at Stake?



Cyberattacks, Quantified

$4.88 million

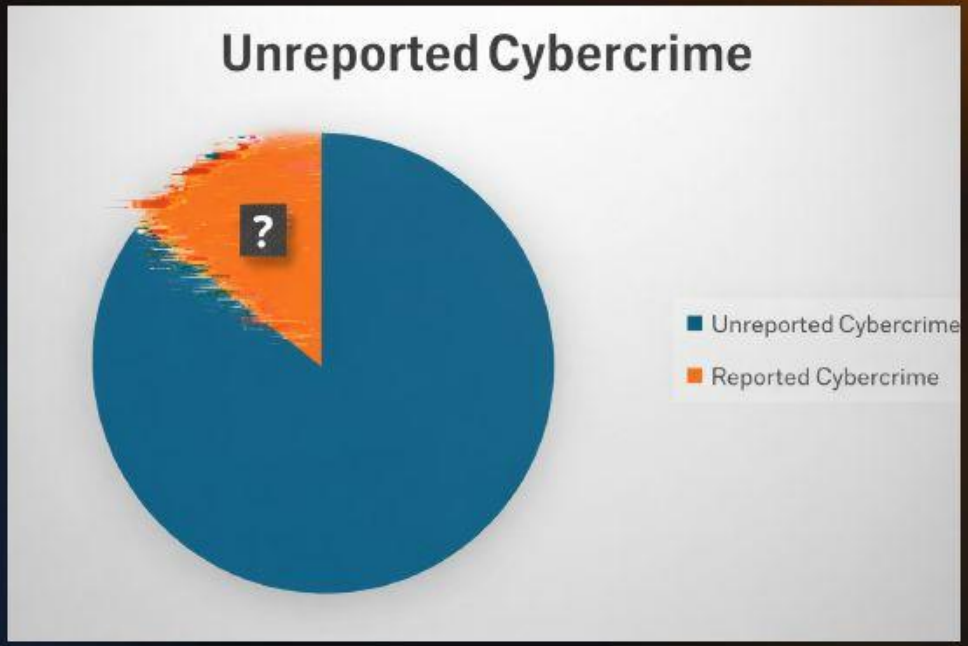
$16.6 billion

$10 billion

19,000+ NHS appointments

The Bottleneck

Dormant Software Trojans



5 Design Requirements

High Detection Accuracy

Low False Positives

Non-intrusive

Scalable

Evasion Resilient

3 Initial Designs

Capability	Static Analysis	Physical Hardware Testing	Firmguard
High Detection Accuracy	❌	✅	✅
Low False Positives	❌	❌	✅
Non-Intrusive	✅	❌	✅
Scalable	✅	❌	✅
Evasion Resilient	❌	❌	✅

Our Solution: Firmguard



Virtual + Parallel Testing

Firmware

Rehost + Parallelize



Digital Twins

Multi-Dimensional Testing

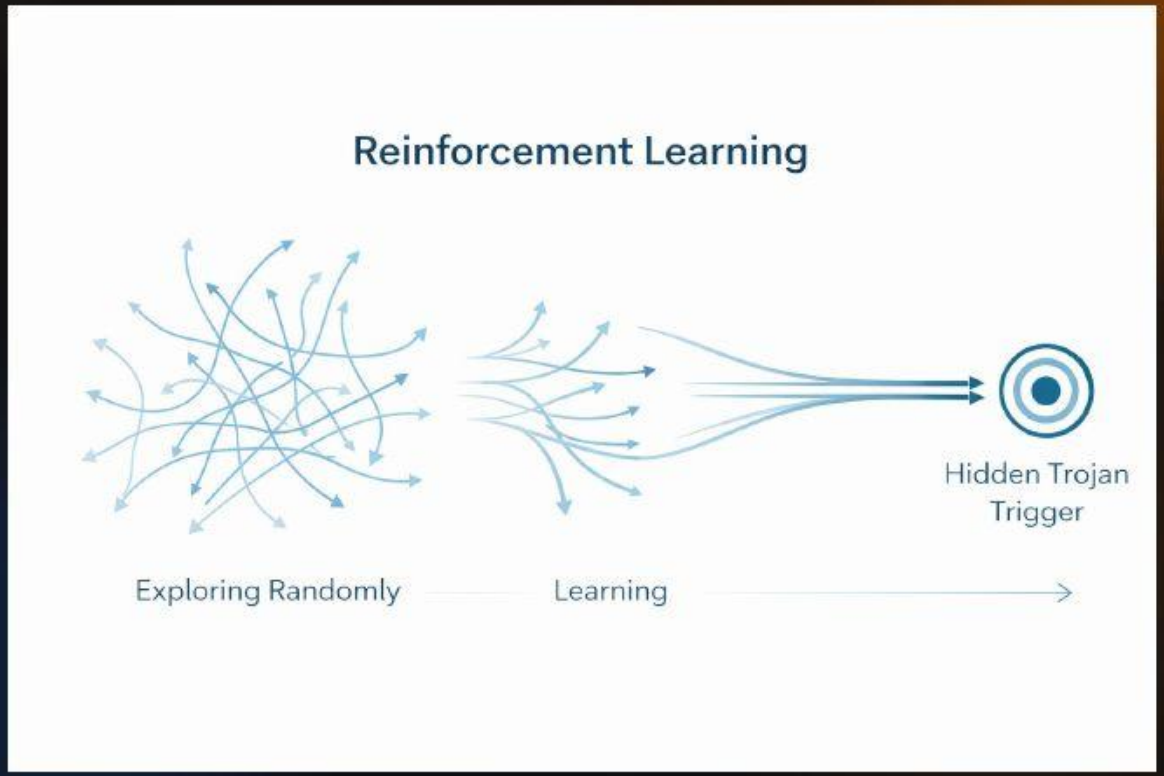
Timing

Inputs

Environment

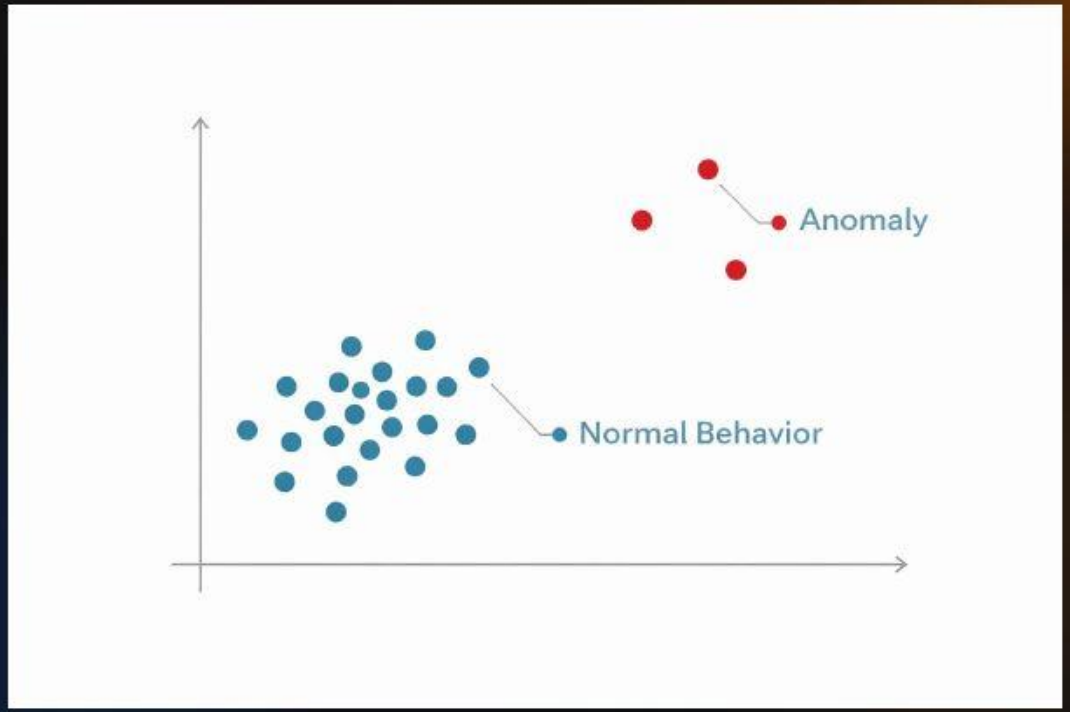
Interrupts
(Events)

AI-Driven Exploration



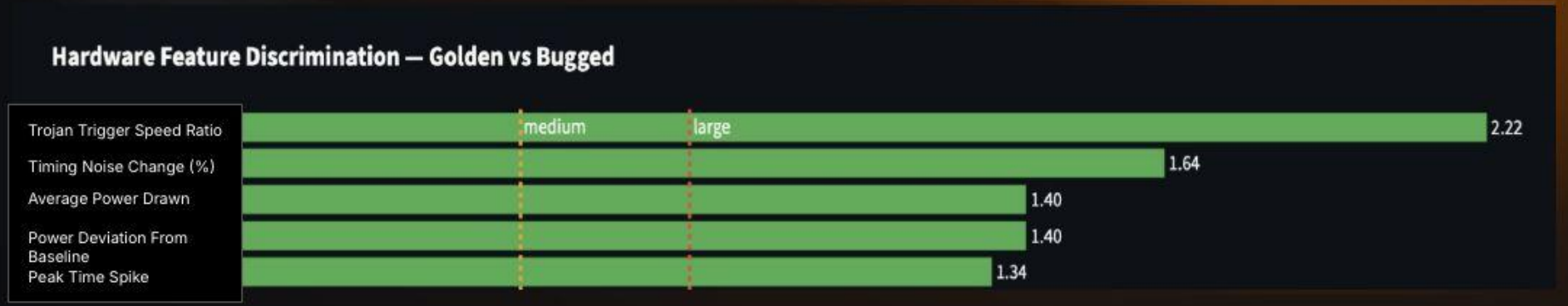
Finding Hidden Anomalies

Power • Timing • CPU • Behavior

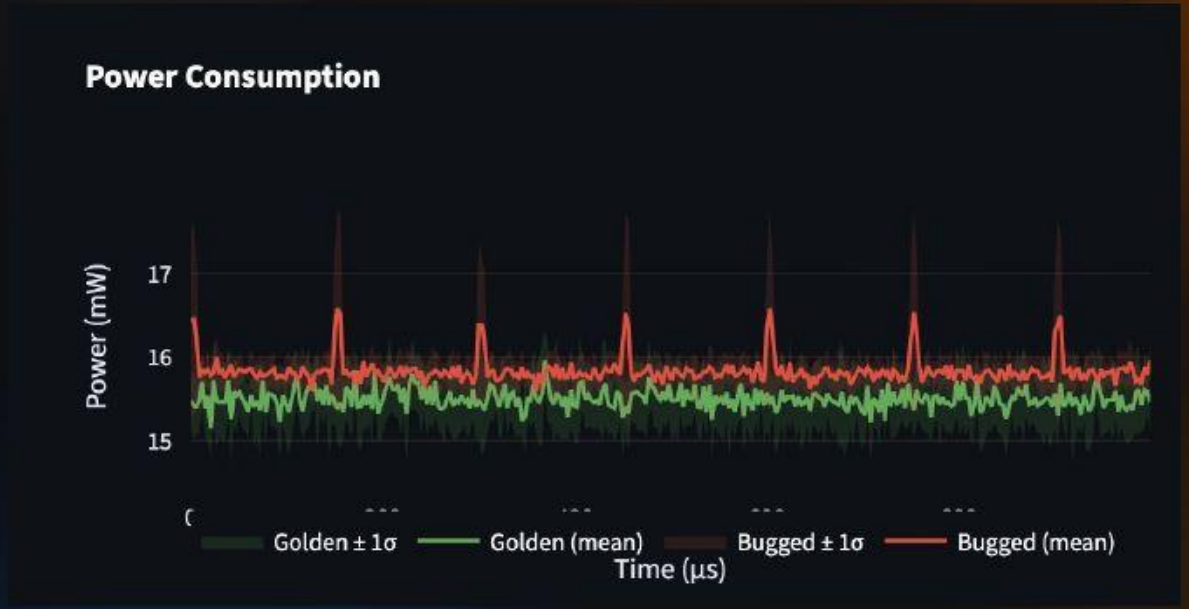


KNN

Hardware Data



Hardware - Data Analysis



Hardware - Data Analysis

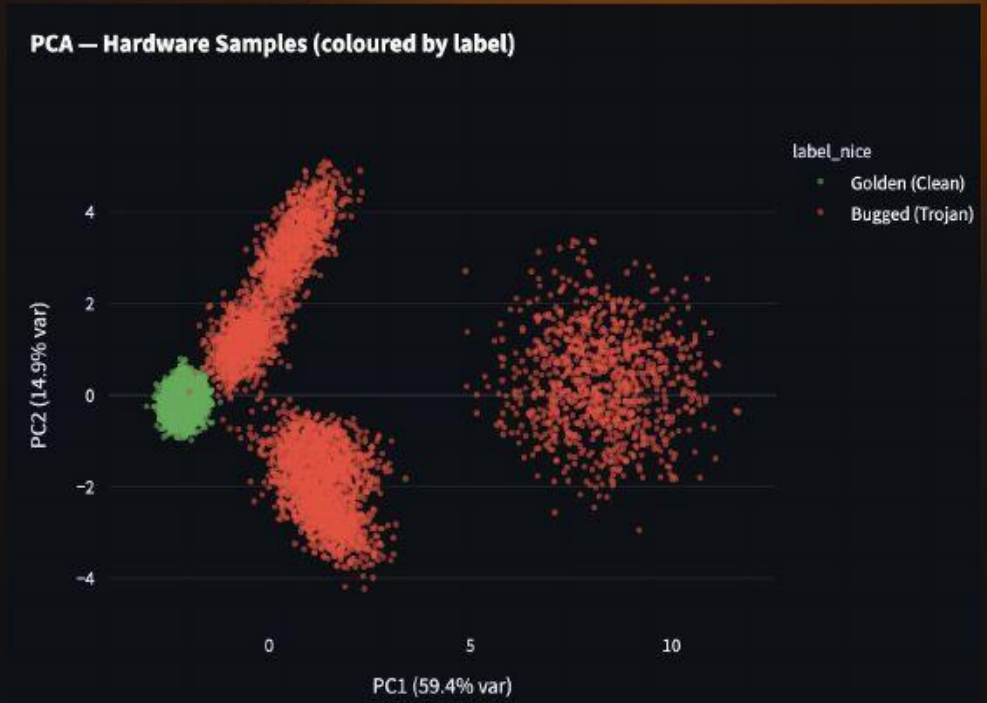


Hardware - Data Analysis

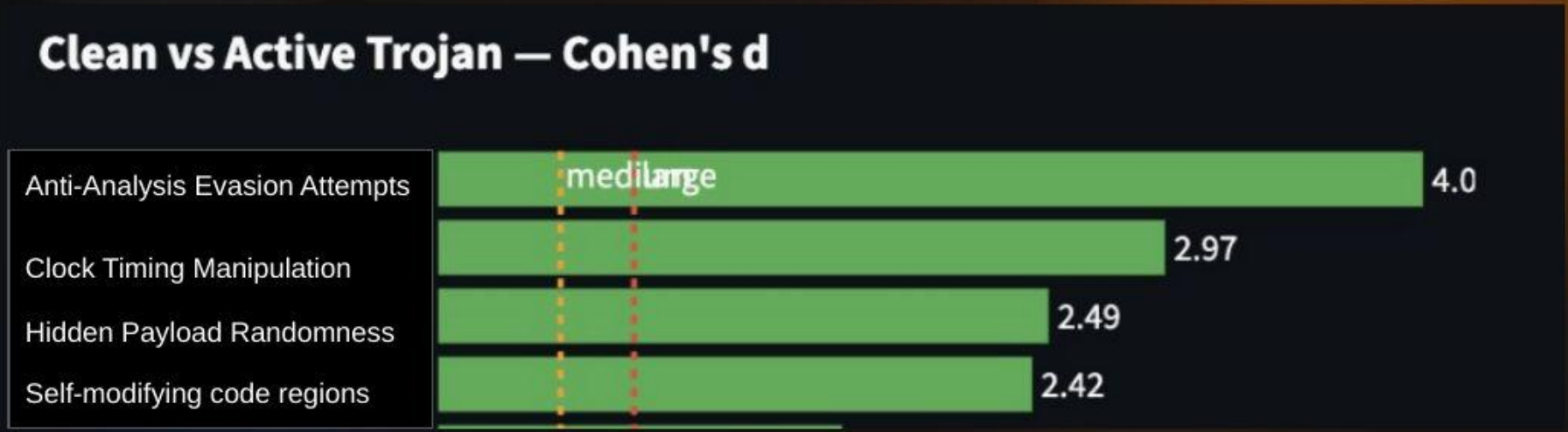


KNN results (hardware)

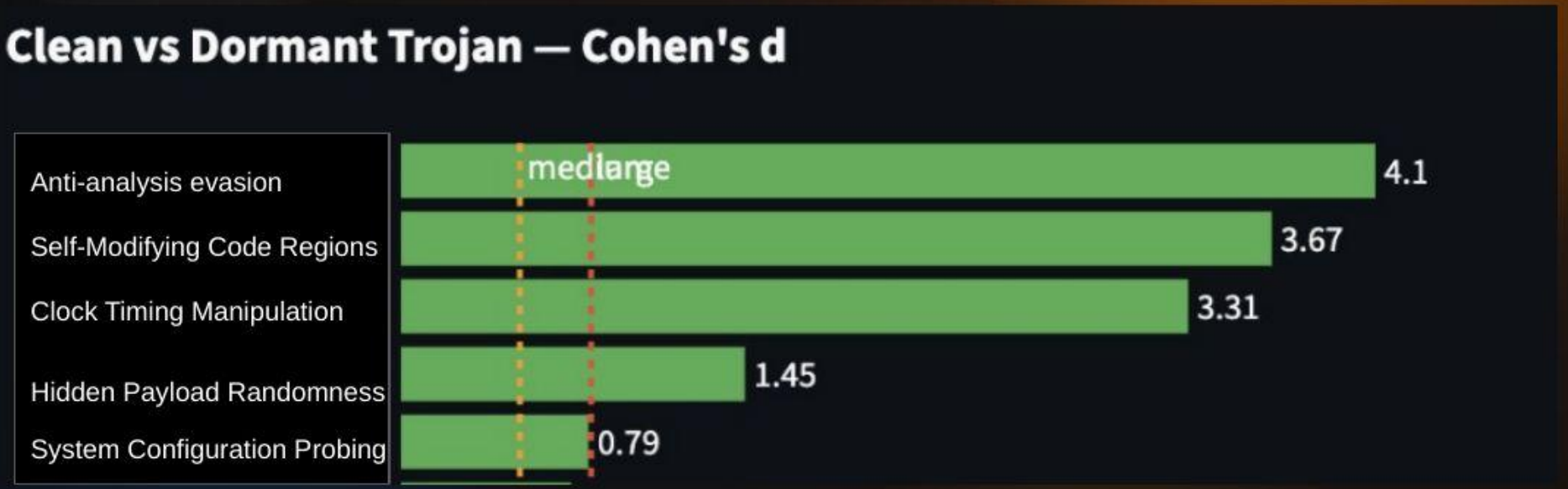




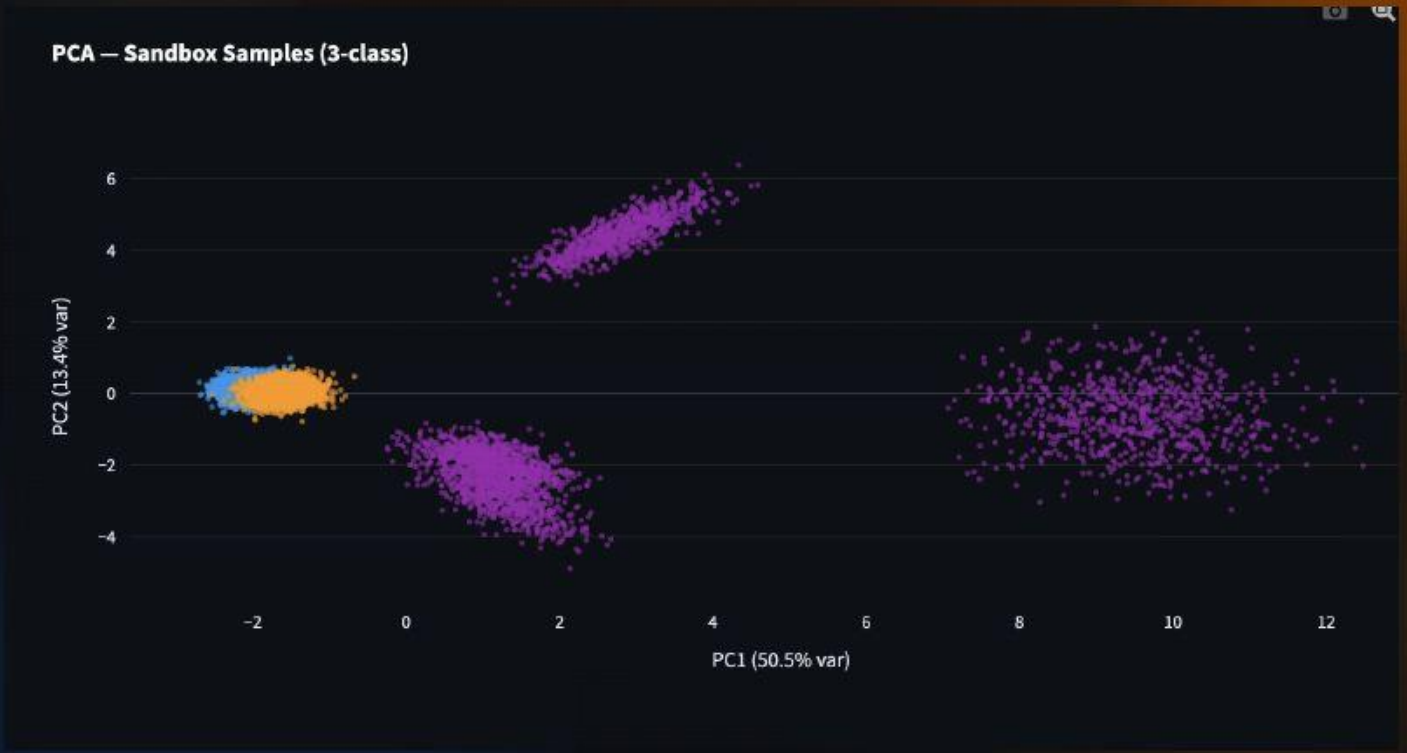
SandBox


Clean vs Active Trojan — Cohen's d
Anti-Analysis Evasion Attempts
Clock Timing Manipulation
Hidden Payload Randomness
Self-modifying code regions
medium
large
4.0
2.97
2.49
2.42

SandBox


Clean vs Dormant Trojan — Cohen's d
Anti-analysis evasion
Self-Modifying Code Regions
Clock Timing Manipulation
Hidden Payload Randomness
System Configuration Probing
medlarge
4.1
3.67
3.31
1.45
0.79

KNN results (SandBox)



Firmguard is…



Efficient



Optimized

Manufacturers

Defense

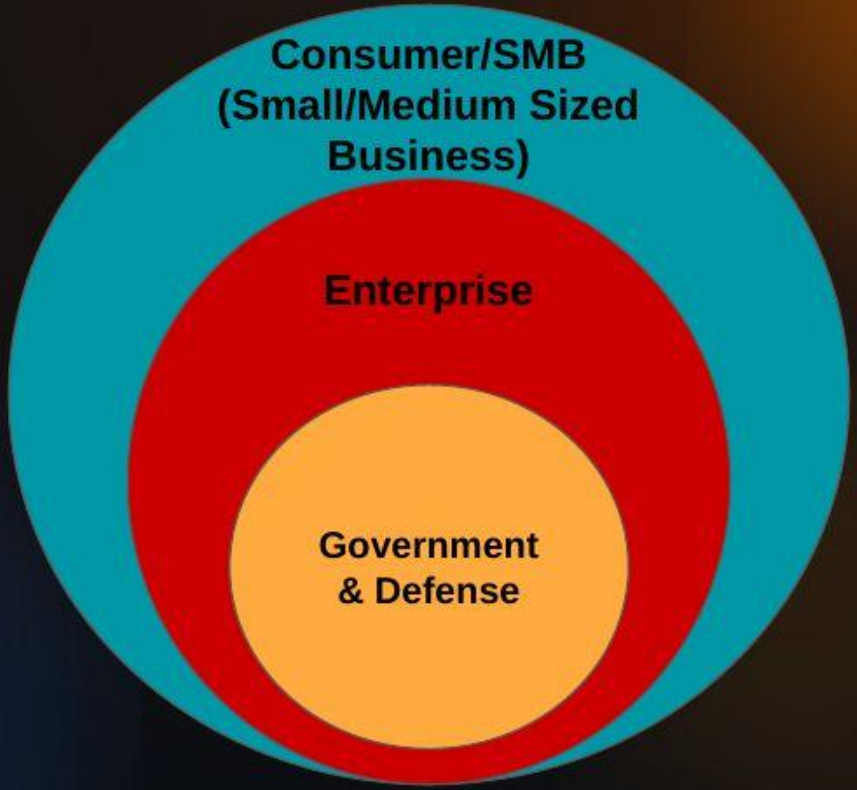
Semiconductors

Critical Infrastructure

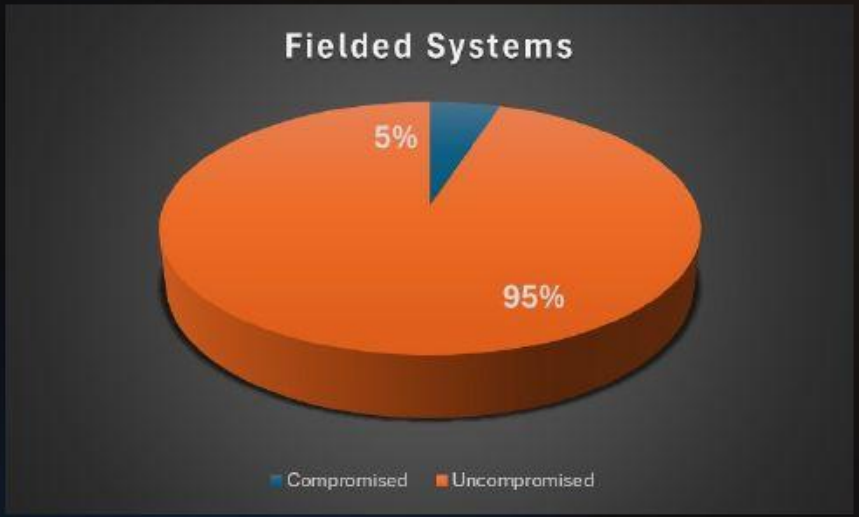


Deployment

Integration



Loss Prevention



Market	Value
TAM	$27.4 Billion
SAM	$5.8 Billion
SOM	$185 Million

Impact Near and Long term



THANK YOU



rehost • trigger • detect

Breakdown of numbers

Market	Value	Description
TAM	**$27.4 Billion**	Everyone globally who could benefit from firmware security screening
SAM	**$5.8 Billion**	The segments we can realistically reach with a SaaS delivery model
SOM	**$185 Million**	What we can capture in years 1–5 across all three tiers

Breakdown of numbers

Segment	TAM Value	Basis
Consumer / SMB IoT Security	$8.2B	Global consumer + SMB IoT security market (2024)
Enterprise Embedded & Supply Chain Security	$14.7B	Enterprise IoT security + embedded systems security + supply chain security combined
Government / Defense Firmware Assurance	$4.5B	US federal cyber spend (firmware-attributable) + allied nation defense embedded programs
Total TAM	**$27.4B**	

Breakdown of numbers

Segment	SAM Value	Basis
Consumer	$1.2B	Device-aware users in US, UK, Canada, Australia who pay for cybersecurity tools
Enterprise	$3.4B	Semiconductor, defense prime, electronics OEM, critical infrastructure in US + EU
Government	$1.2B	DoD, DHS, intelligence community, Five Eyes allies, NATO programs
Total SAM	**$5.8B**	

Breakdown of numbers

Breakdown

Segment	SOM Value	Assumptions
Consumer	$15M	150,000 subscribers × $99/year
Enterprise	$120M	600 enterprise contracts × $200,000 average annual contract value
Government	$50M	5 federal/defense contracts × $10M average
Total SOM	**$185M**	

Year-by-Year Ramp

Year	Revenue Target	Primary Driver
Year 1	$2M	Pilot enterprise contracts, early consumer beta
Year 2	$12M	First government SBIR award, enterprise expansion
Year 3	$45M	DoD contract, 100+ enterprise clients
Year 4	$105M	FedRAMP certification, NATO/Five Eyes entry
Year 5	$185M	Full three-tier commercial operation